Test Report

Fail

Test Configuration Details	
Application	
Name	D9020DPHC MIPI D-PHY Test
Version	4.0.24.0
Device Description	
High Speed Data Rate(Mbps)	4500
CTS Version	v2.0 and v2.1
ZID	100 ohm
CLoad	50pF
Fixture Setup	Auto Load Switching
Test Session Details	
Infiniium SW Version	06.74.00701
Infiniium Model Number	DSOX92504A
Infiniium Serial Number	MY51420103
Debug Mode Used	No
Compliance Limits	MIPI D-PHY Test Limit v2.0 and v2.1 (official)
Probe (Channel 1)	Model: MX0021A Serial: US61180146 Head: MX0100A Atten: Not Calibrated, Using Default Atten (7.6823E+00) Skew: Not Calibrated, Using Default Skew
Probe (Channel 2)	Model: MX0021A Serial: US61180121 Head: MX0100A Atten: Not Calibrated, Using Default Atten (7.6220E+00) Skew: Not Calibrated, Using Default Skew
Probe (Channel 3)	Model: MX0021A Serial: US61180140 Head: MX0100A Atten: Not Calibrated, Using Default Atten (7.5690E+00) Skew: Not Calibrated, Using Default Skew
Probe (Channel 4)	Model: MX0021A Serial: US61180136 Head: MX0100A Atten: Not Calibrated, Using Default Atten (7.6519E+00) Skew: Not Calibrated, Using Default Skew
Last Test Date	2025-07-28 14:46:24 UTC +08:00

Summary of Results

Test Statistics	
Failed	4
Passed	7
Total	11

Margin Thresholds	
Warning	< 5 %
Critical	< 0 %

Pass	# Failed	# Trials	Test Name (click to jump)	Actual Value	Margin	Pass Limits
✔	0	1	1.2.1 ULPS Clock TX Thevenin Output High Voltage Level (VOH) ULPSMODE	1.23700 V	18.0 %	LPThevenin_VOH_Limit_min V <= VALUE <= 1.30000 V
✔	0	1	1.2.2 ULPS Clock TX Thevenin Output Low Voltage Level (VOL) ULPSMODE	38.00 mV	12.0 %	-50.00 mV <= VALUE <= 50.00 mV
✘	1	1	1.2.3 ULPS Clock TX 15%-85% Rise Time (TRLP) ULPSMODE	31.04 ns	-24.2 %	VALUE < 25.00 ns
✘	1	1	1.2.4 ULPS Clock TX 15%-85% Fall Time (TFLP) ULPSMODE	30.38 ns	-21.5 %	VALUE < 25.00 ns
✔	0	1	1.1.1 LP TX Thevenin Output High Voltage Level (VOH) ESCAPEMODE	1.270 V	8.6 %	LPThevenin_VOH_Limit_min V <= VALUE <= 1.300 V
✔	0	1	1.1.2 LP TX Thevenin Output Low Voltage Level (VOL) ESCAPEMODE	23.00 mV	27.0 %	-50.00 mV <= VALUE <= 50.00 mV
✘	1	1	1.1.3 LP TX 15%-85% Rise Time (TRLP) ESCAPEMODE	32.07 ns	-28.3 %	VALUE < 25.00 ns
✘	1	1	1.1.4 LP TX 15%-85% Fall Time (TFLP) ESCAPEMODE	31.80 ns	-27.2 %	VALUE < 25.00 ns
✔	0	1	1.1.6 LP TX Pulse Width of LP TX Exclusive-OR Clock (TLP-PULSE-TX)	141.91 ns	609.6 %	VALUE > 20.00 ns
✔	0	1	1.1.6 LP TX Pulse Width of LP TX Exclusive-OR Clock (TLP-PULSE-TX) [Initial]	154.51 ns	286.3 %	VALUE > 40.00 ns
✔	0	1	1.1.7 LP TX Period of LP TX Exclusive-OR Clock (TLP-PER-TX)	301.82 ns	235.4 %	VALUE > 90.00 ns

Report Detail

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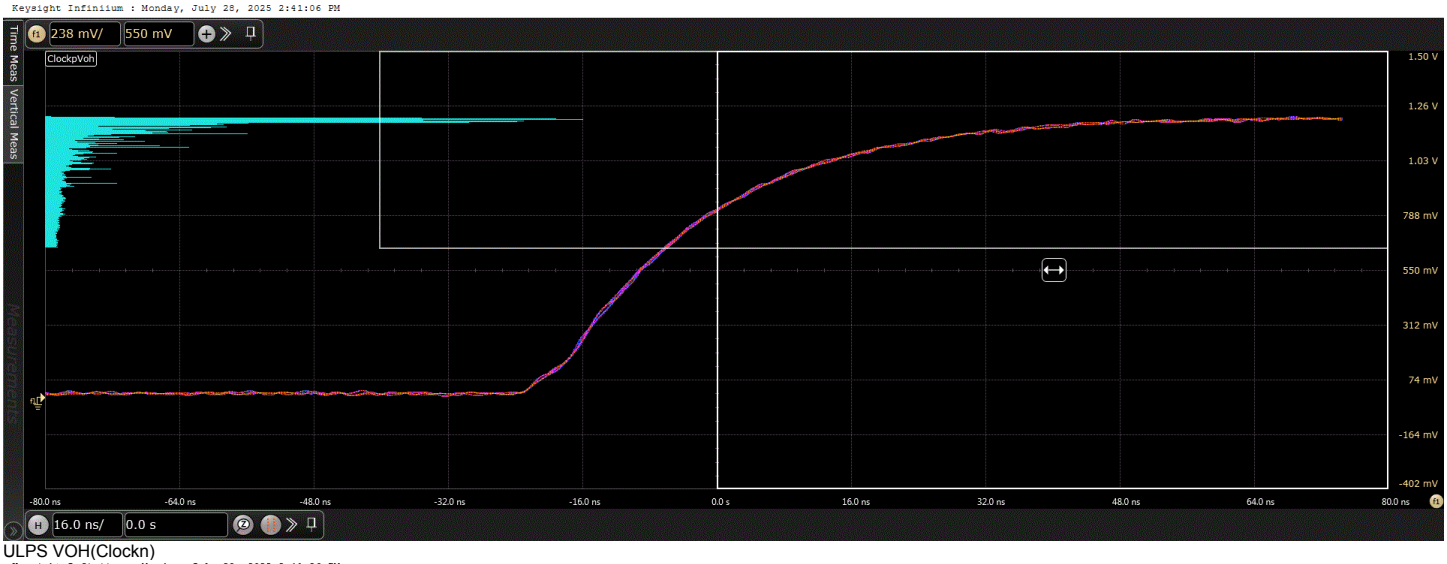
✔ 1.2.1 ULPS Clock TX Thevenin Output High Voltage Level (VOH) ULPSMODE

D-PHY Specification v2.1 Section 9.1.2 Table 27

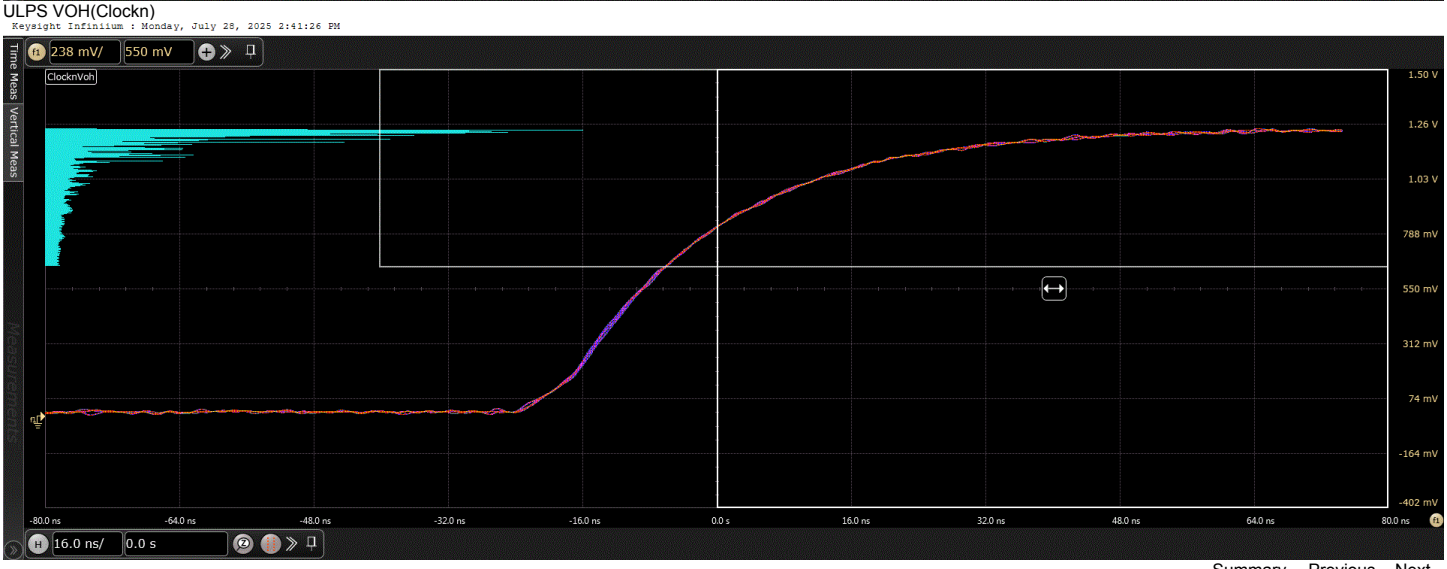
This test is to verify that the Thevenin Output High Voltage of clock lane using ULPS mode is between the minimum conformance limit(LPThevenin_VOH_Limit_min) and the maximum conformance limit of 1.3V.
For CTS v1.0 and CTS v1.1, the minimum conformance limit(LPThevenin_VOH_Limit_min) is 1.1V.
For CTS v1.2, the minimum conformance limit(LPThevenin_VOH_Limit_min) is 1.1V for Datarate = 1.5Gbps, 0.95V for Datarate > 1.5Gbps.
Actual Value Measurement Name: 28211_Clock VOH(Worst)
Pass Limits: LPThevenin_VOH_Limit_min V <= VALUE <= 1.30000 V

Actual Value	Margin	ULPS VOH(Clockp)	ULPS VOH(Clockn)	Histogram Result	ClkpVoh	ClknVoh	CLoad	PassLimit Min (LPThevenin_VOH_Limit_min)
1.23700 V	18.0 %	(See image)	(See image)	Mode	1.207 V	1.237 V	50pF	950.00 mV

ULPS VOH(Clockp)



ULPS VOH(Clockn)



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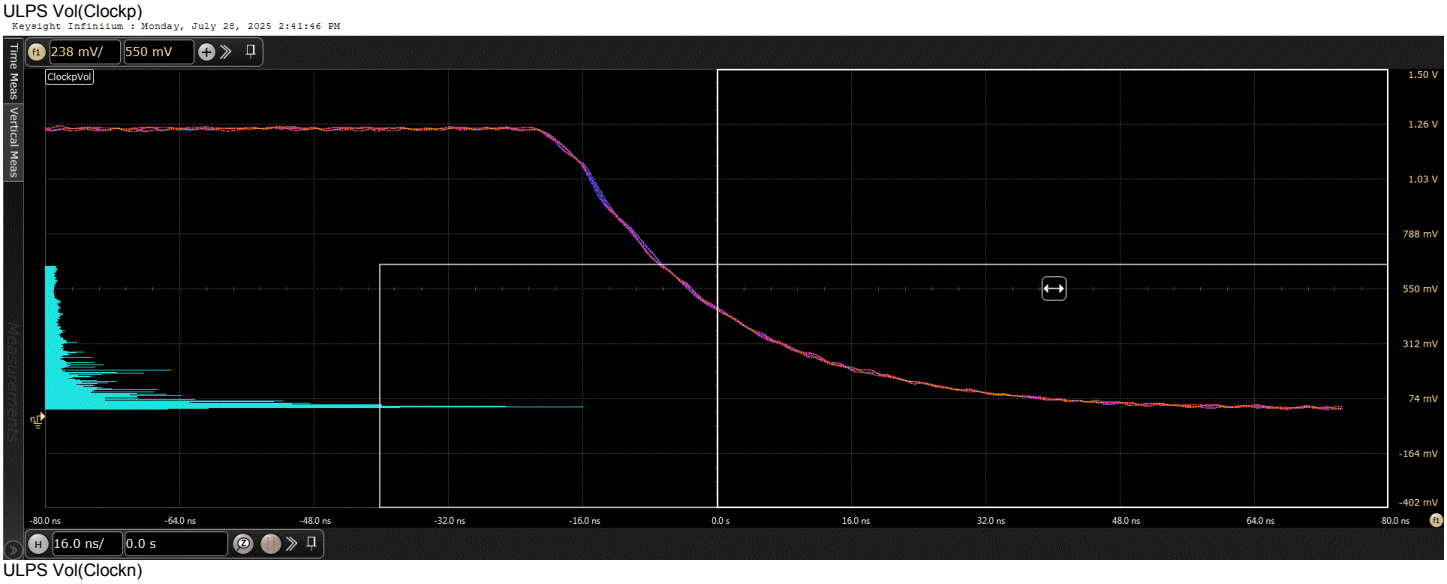
1.2.2 ULPS Clock TX Thevenin Output Low Voltage Level (VOL) ULPSMODE

D-PHY Specification v2.1 Section 9.1.2 Table 27

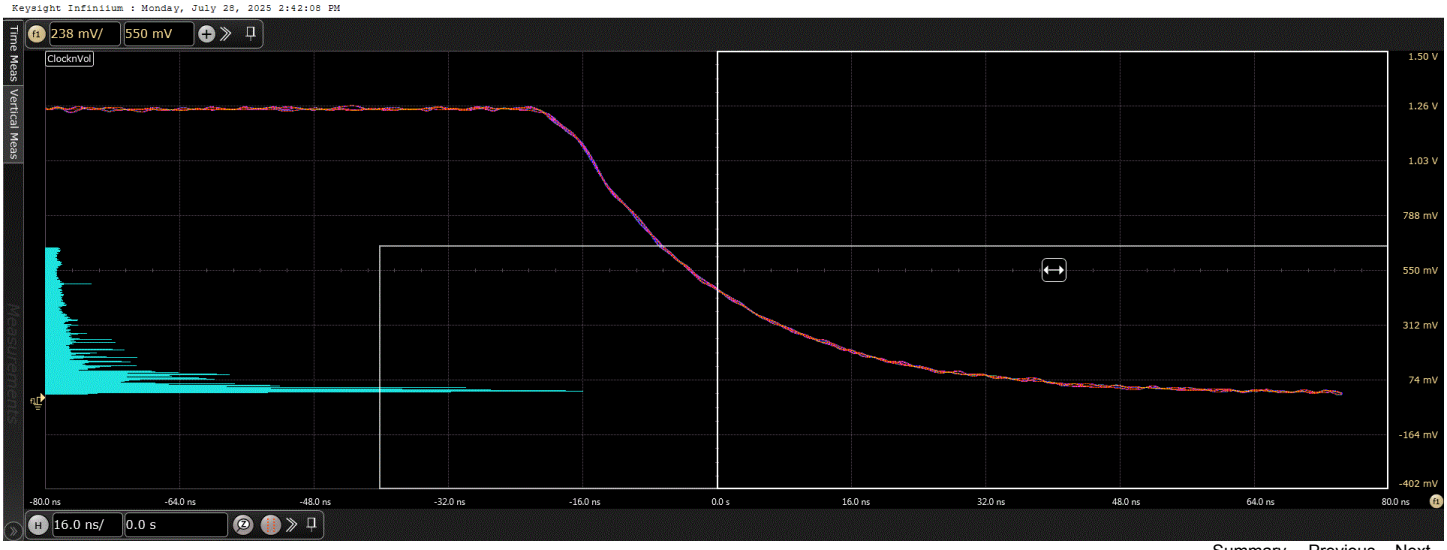
This test is to verify that the Thevenin Output Low Voltage of clock lane using ULPS mode is within the conformance limits.
Actual Value Measurement Name: 28221_Clock VOL(Worst)
Pass Limits: -50.00 mV <= VALUE <= 50.00 mV

Actual Value	Margin	ULPS Vol(Clockp)	ULPS Vol(Clockn)	Histogram Result	ClkpVol	ClknVol	CLoad
38.00 mV	12.0 %	(See image)	(See image)	Mode	38.00 mV	25.00 mV	50pF

ULPS Vol(Clockp)



ULPS Vol(Clockn)



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1.2.3 ULPS Clock TX 15%-85% Rise Time (TRLP) ULPSMODE

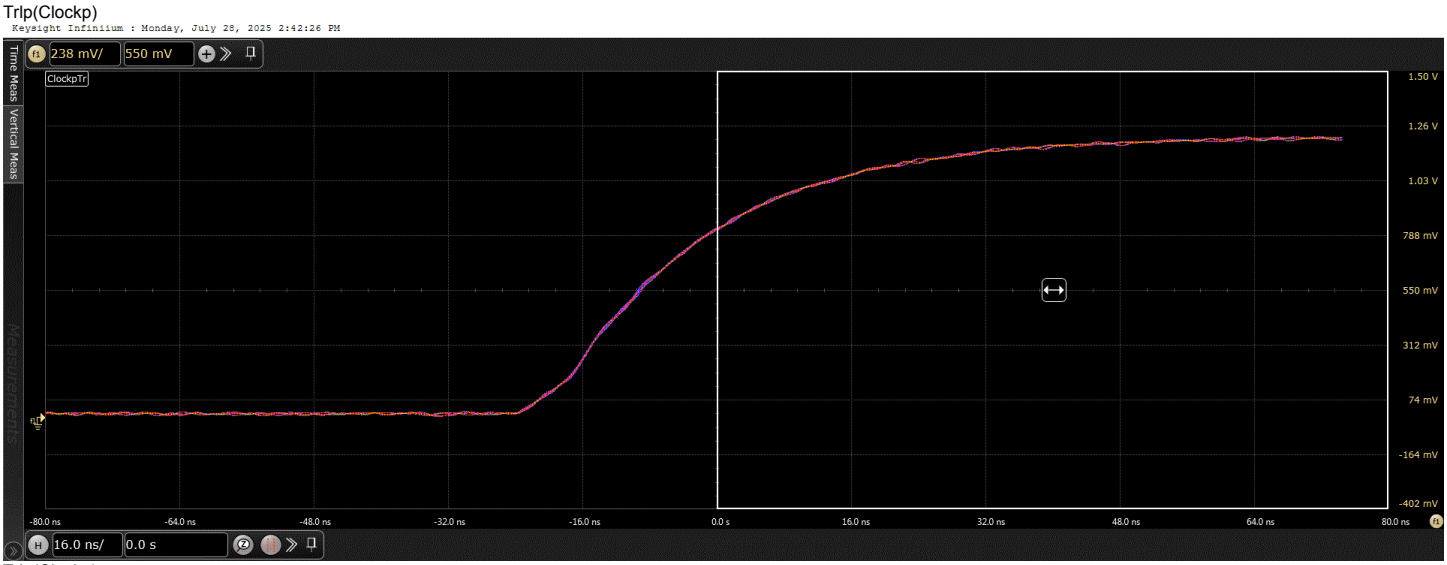
D-PHY Specification v2.1 Section 9.1.2 Table 28

15%-85% rise time of LP signal on clock lane using Escape Mode.
Actual Value Measurement Name: 28241_Clock Trlp(Worst)
Pass Limits: VALUE < 25.00 ns

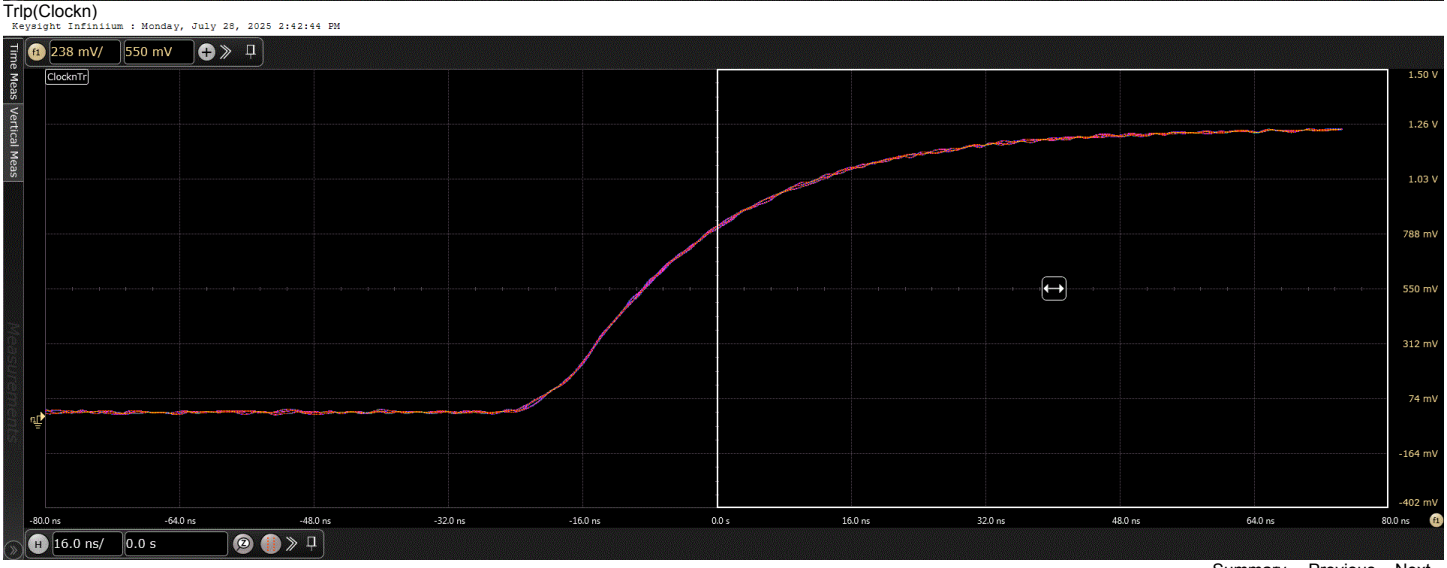
Actual Value	Margin	Trlp(Clockp)	Trlp(Clockn)	ClkpTrlp(Mean)	ClkpTrlp(Max)	ClkpTrlp(Min)	Clkp LP Edge Measured:	ClknTrlp(Mean)
31.04 ns	-24.2 %	(See image)	(See image)	30.81 ns	31.05 ns	30.43 ns	(no value)	31.04 ns

ClknTrlp(Max)	ClknTrlp(Min)	Clkn LP Edge Measured:	CLoad
31.53 ns	30.78 ns	(no value)	50pF

Trlp(Clockp)



Trlp(Clockn)



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1.2.4 ULPS Clock TX 15%-85% Fall Time (TFLP) ULPSMODE

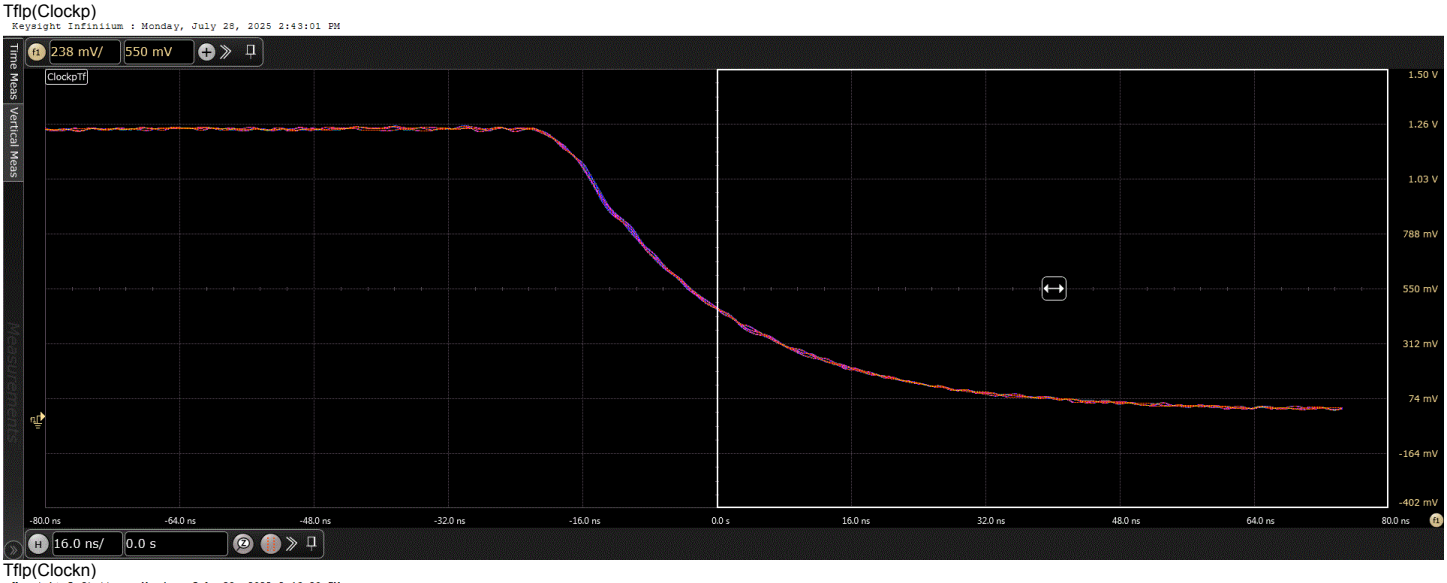
D-PHY Specification v2.1 Section 9.1.2 Table 28

15%-85% fall time of LP signal for clock lane using Escape Mode.
Actual Value Measurement Name: 28251_Clock Tflp(Worst)
Pass Limits: VALUE < 25.00 ns

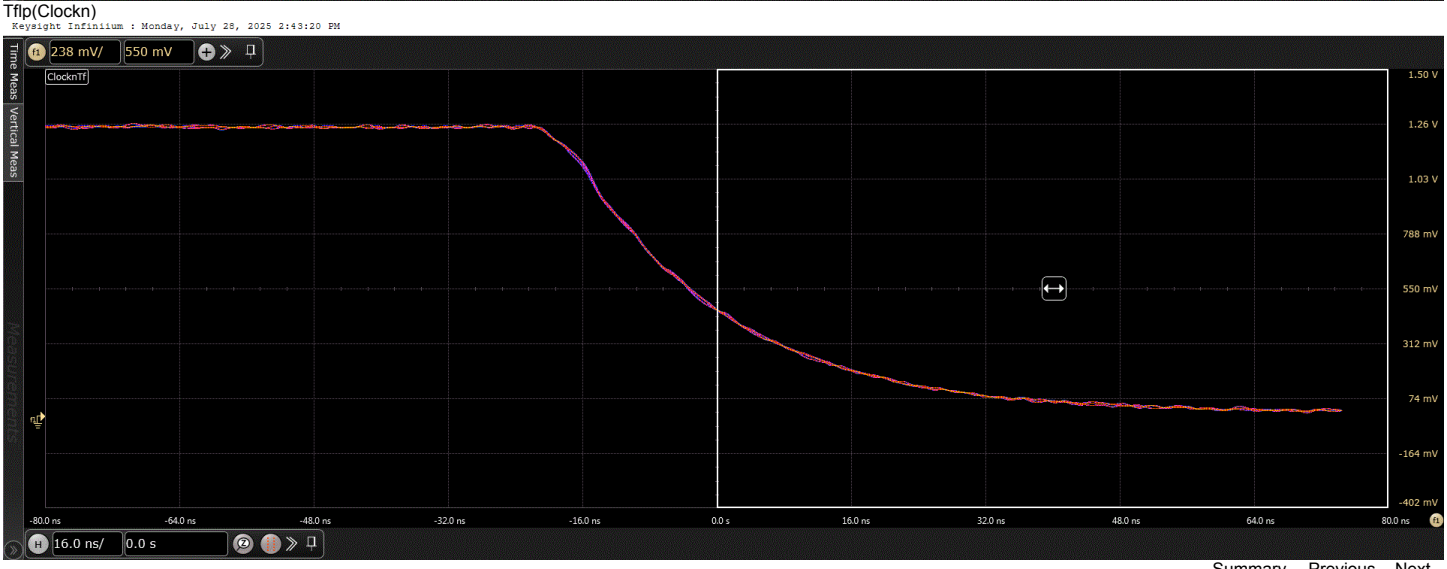
Actual Value	Margin	Tflp(Clockp)	Tflp(Clockn)	ClkpTflp(Mean)	ClkpTflp(Max)	ClkpTflp(Min)	Clkp LP Edge Measured:	ClknTflp(Mean)
30.38 ns	-21.5 %	(See image)	(See image)	30.23 ns	30.46 ns	29.94 ns	(no value)	30.38 ns

ClknTflp(Max)	ClknTflp(Min)	Clkn LP Edge Measured:	CLoad
30.66 ns	30.22 ns	(no value)	50pF

Tflp(Clockp)



Tflp(Clockn)



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1.1.1 LP TX Thevenin Output High Voltage Level (VOH) ESCAPEMODE

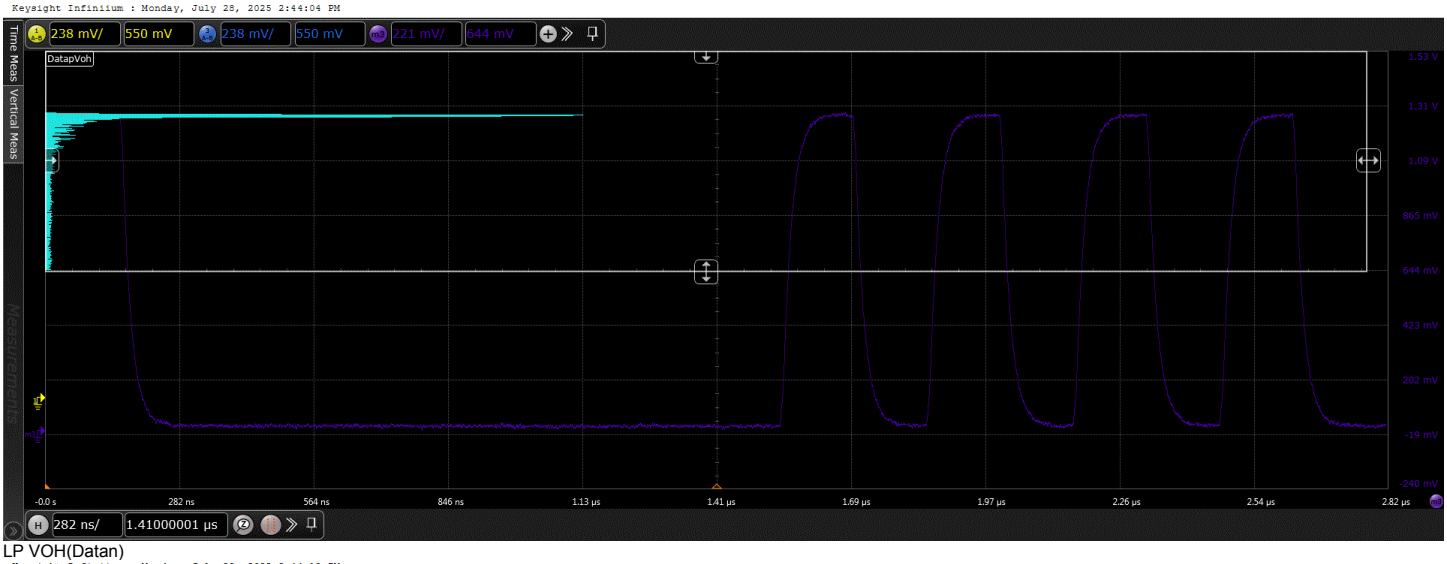
D-PHY Specification v2.1 Section 9.1.2 Table 27

This test is to verify that the Thevenin Output High Voltage of data lane using escape mode is between the minimum conformance limit(LPThevenin_VOH_Limit_min) and the maximum conformance limit of 1.3V.
For CTS v1.0 and CTS v1.1, the minimum conformance limit(LPThevenin_VOH_Limit_min) is 1.1V.
For CTS v1.2, the minimum conformance limit(LPThevenin_VOH_Limit_min) is 1.1V for Datarate = 1.5Gbps, 0.95V for Datarate > 1.5Gbps.
Actual Value Measurement Name: 8211_VOH(Worst)
Pass Limits: LPThevenin_VOH_Limit_min V <= VALUE <= 1.300 V

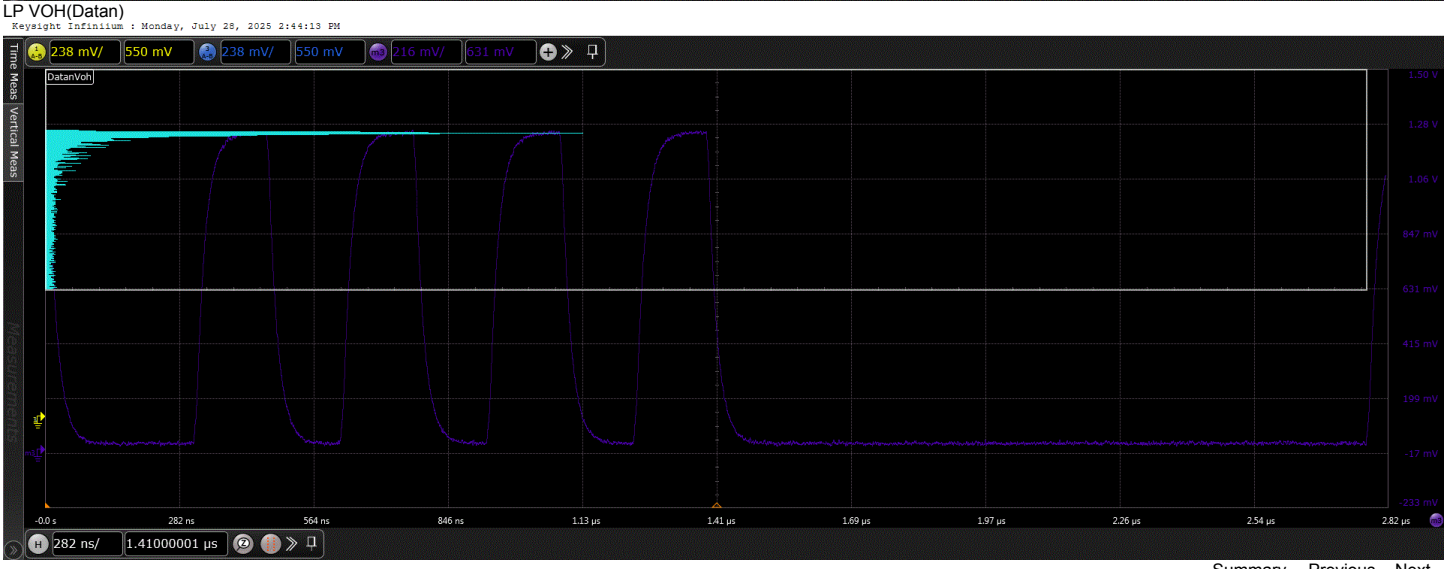
Actual Value	Margin	LP VOH(Datap)	LP VOH(Datan)	Histogram Result	DpVoh	DnVoh	CLoad	DataLane
1.270 V	8.6 %	(See image)	(See image)	Mode	1.270 V	1.243 V	50pF	Lane0

PassLimit Min (LPThevenin_VOH_Limit_min)
950 mV

LP VOH(Datap)



LP VOH(Datan)



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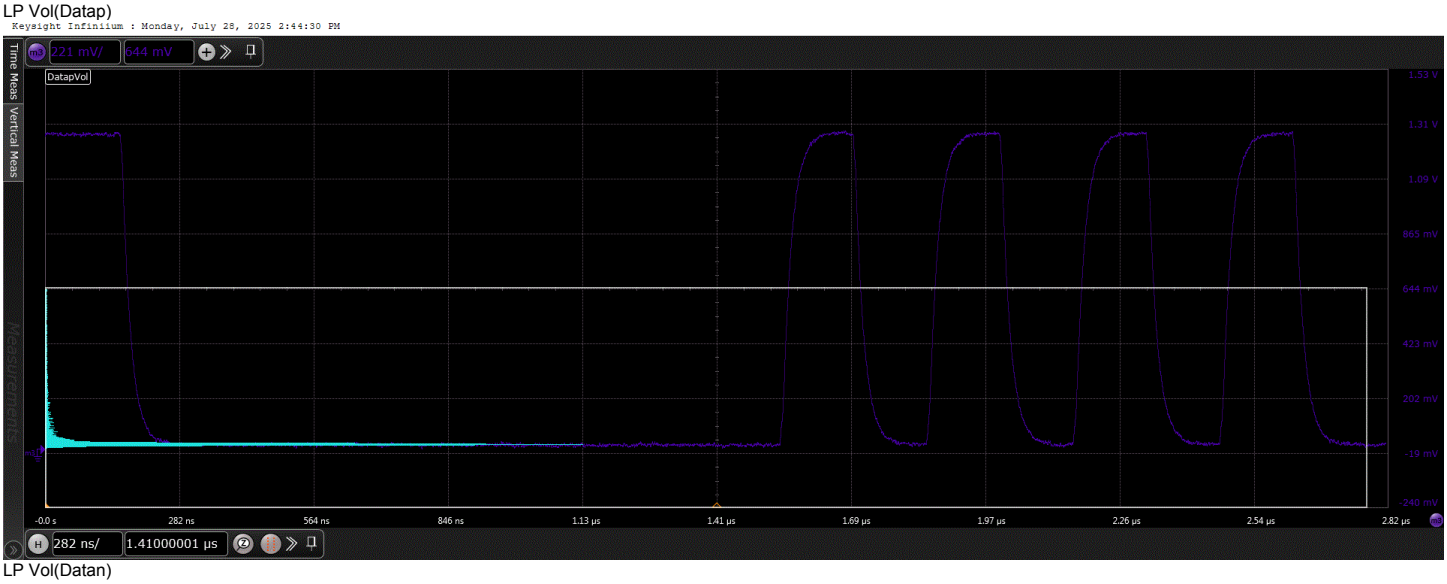
1.1.2 LP TX Thevenin Output Low Voltage Level (VOL) ESCAPEMODE

D-PHY Specification v2.1 Section 9.1.2 Table 27

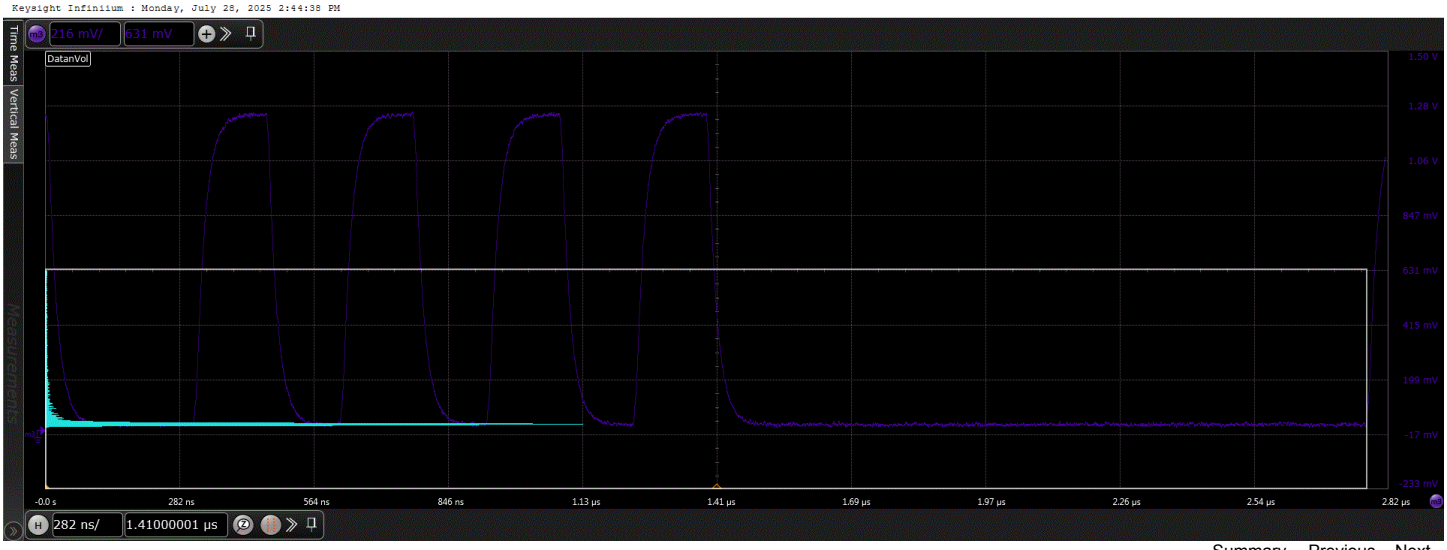
This test is to verify that the Thevenin Output Low Voltage of data lane using escape mode is within the conformance limits.
Actual Value Measurement Name: 8221_VOL(Worst)
Pass Limits: -50.00 mV <= VALUE <= 50.00 mV

Actual Value	Margin	LP Vol(Datap)	LP Vol(Datan)	Histogram Result	DpVol	DnVol	CLoad	DataLane
23.00 mV	27.0 %	(See image)	(See image)	Mode	16.00 mV	23.00 mV	50pF	Lane0

LP Vol(Datap)



LP Vol(Datan)



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1.1.3 LP TX 15%-85% Rise Time (TRLP) ESCAPEMODE

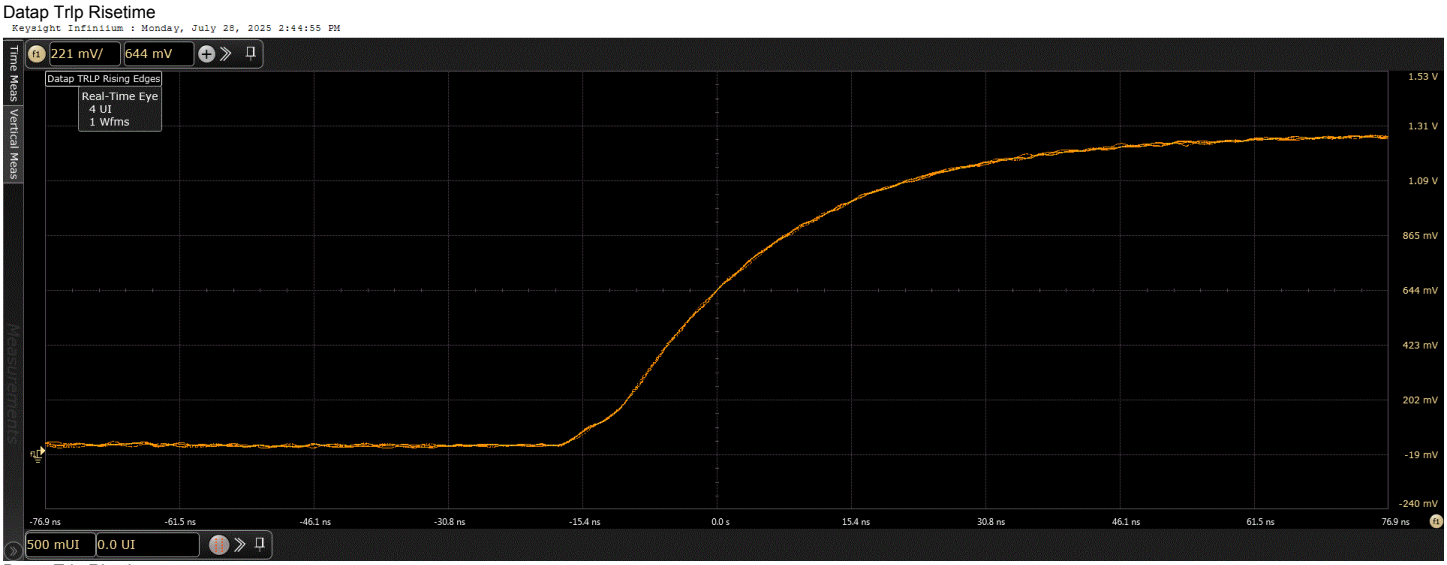
D-PHY Specification v2.1 Section 9.1.2 Table 28

15%-85% rise time of LP signal. (Escape Mode)
Actual Value Measurement Name: 8241_Trlp(Worst)
Pass Limits: VALUE < 25.00 ns

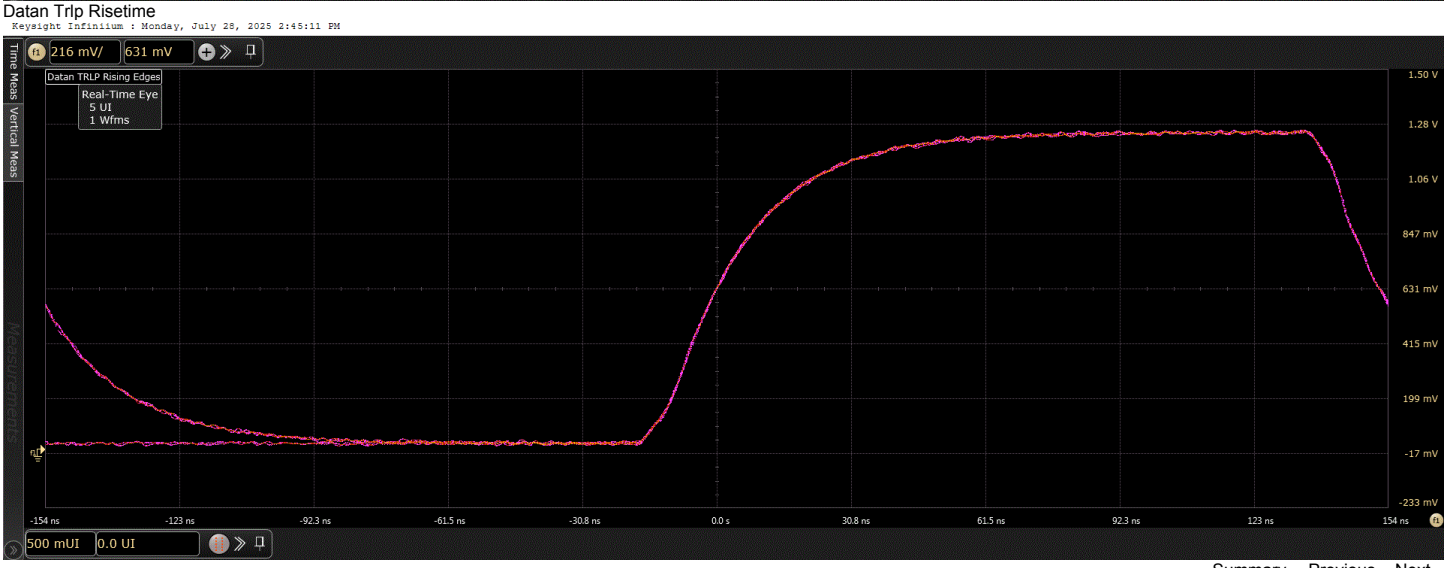
Actual Value	Margin	Datap Trlp Risetime	Datan Trlp Risetime	DpTrlp(Mean)	DpTrlp(Max)	DpTrlp(Min)	Dp LP Edge Measured:	DnTrlp(Mean)
32.07 ns	-28.3 %	(See image)	(See image)	32.07 ns	32.78 ns	31.54 ns	4	31.81 ns

DnTrlp(Max)	DnTrlp(Min)	Dn LP Edge Measured:	CLoad	DataLane
32.12 ns	31.30 ns	5	50pF	Lane0

Datap Trlp Risetime



Datan Trlp Risetime



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1.1.4 LP TX 15%-85% Fall Time (TFLP) ESCAPEMODE

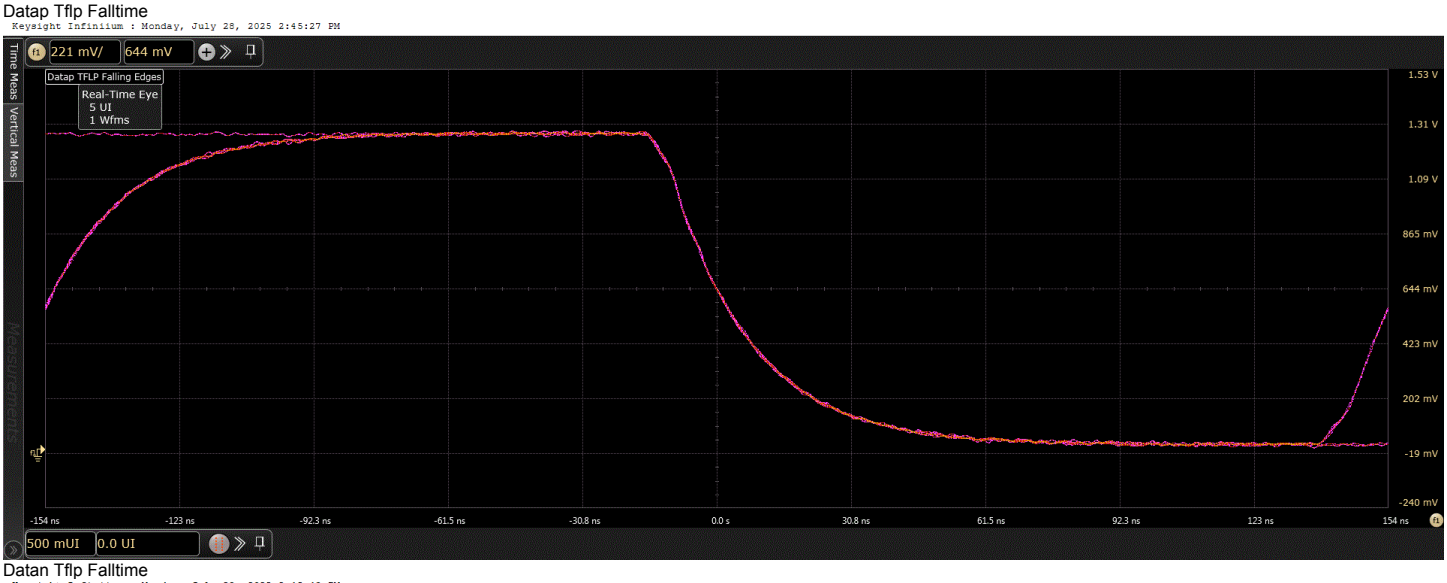
D-PHY Specification v2.1 Section 9.1.2 Table 28

15%-85% fall time of LP signal in Escape Mode.
Actual Value Measurement Name: 8251_Tflp(Worst)
Pass Limits: VALUE < 25.00 ns

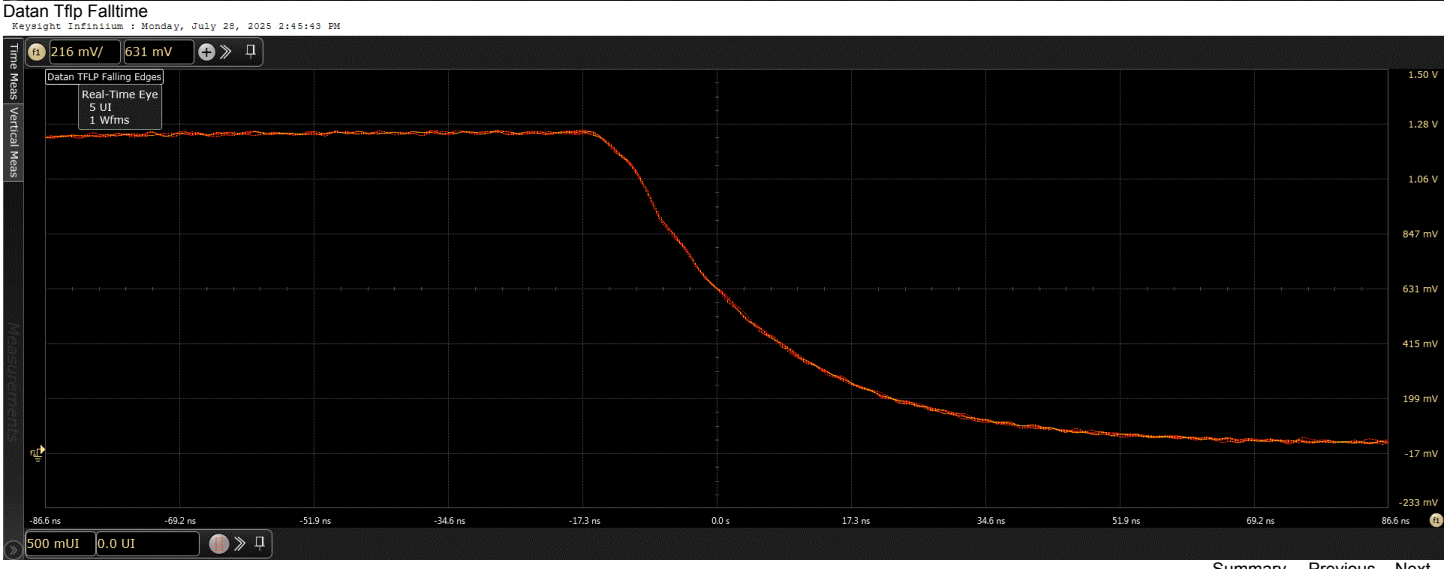
Actual Value	Margin	Datap Tflp Falltime	Datan Tflp Falltime	DpTflp(Mean)	DpTflp(Max)	DpTflp(Min)	Dp LP Edge Measured:	DnTflp(Mean)
31.80 ns	-27.2 %	(See image)	(See image)	31.80 ns	32.16 ns	31.55 ns	5	31.51 ns

DnTflp(Max)	DnTflp(Min)	Dn LP Edge Measured:	CLoad	DataLane
31.88 ns	30.98 ns	5	50pF	Lane0

Datap Tflp Falltime



Datan Tflp Falltime



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1.1.6 LP TX Pulse Width of LP TX Exclusive-OR Clock (TLP-PULSE-TX)

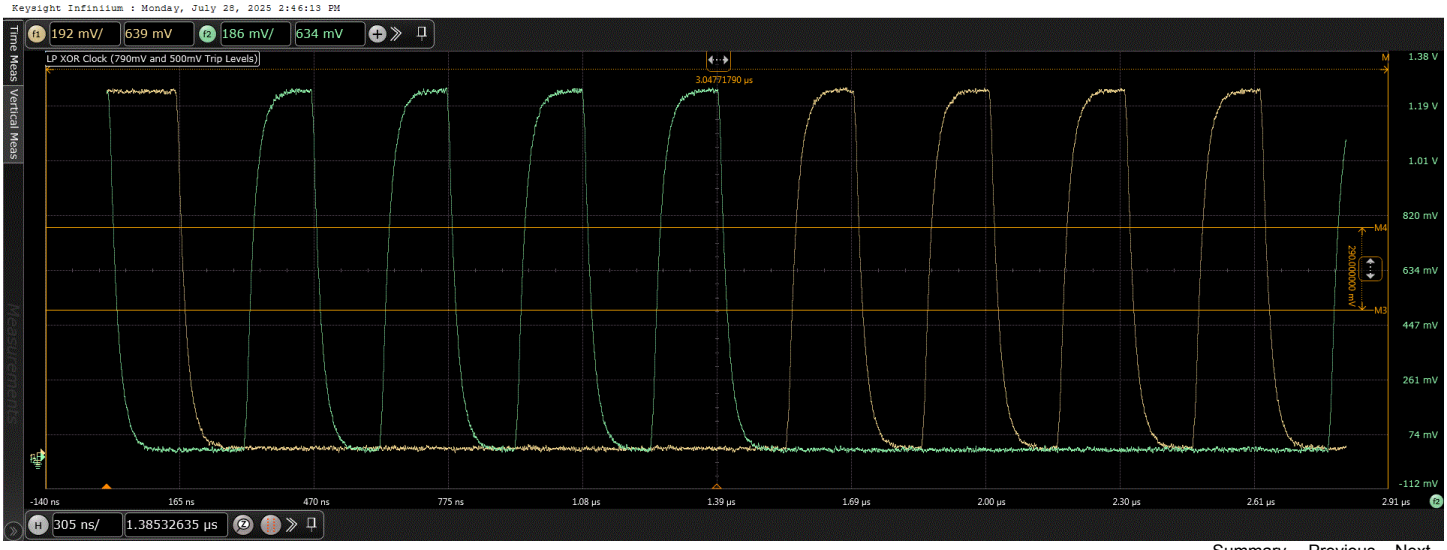
D-PHY Specification v2.1 Section 9.1.2 Table 28

Pulse width measurement for all pulses of the LP exclusive-OR clock excluding the first pulse and last pulse.
Actual Value Measurement Name: 827_Min Pulse Width of LP XOR Clock(Worst)
Pass Limits: VALUE > 20.00 ns

Actual Value	Margin	Max Trip-level Threshold Voltage	Min Pulse Width of LP XOR Clock@Max Trip-level Threshold Voltage
141.91 ns	609.6 %	790.00 mV	141.91 ns

Min Pulse Width of LP XOR Clock@500mV	CLoad	DataLane	TLP-PULSE-TX Screen Capture
159.63 ns	50pF	Lane0	(See image)

TLP-PULSE-TX Screen Capture



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1.1.6 LP TX Pulse Width of LP TX Exclusive-OR Clock (TLP-PULSE-TX) [Initial]

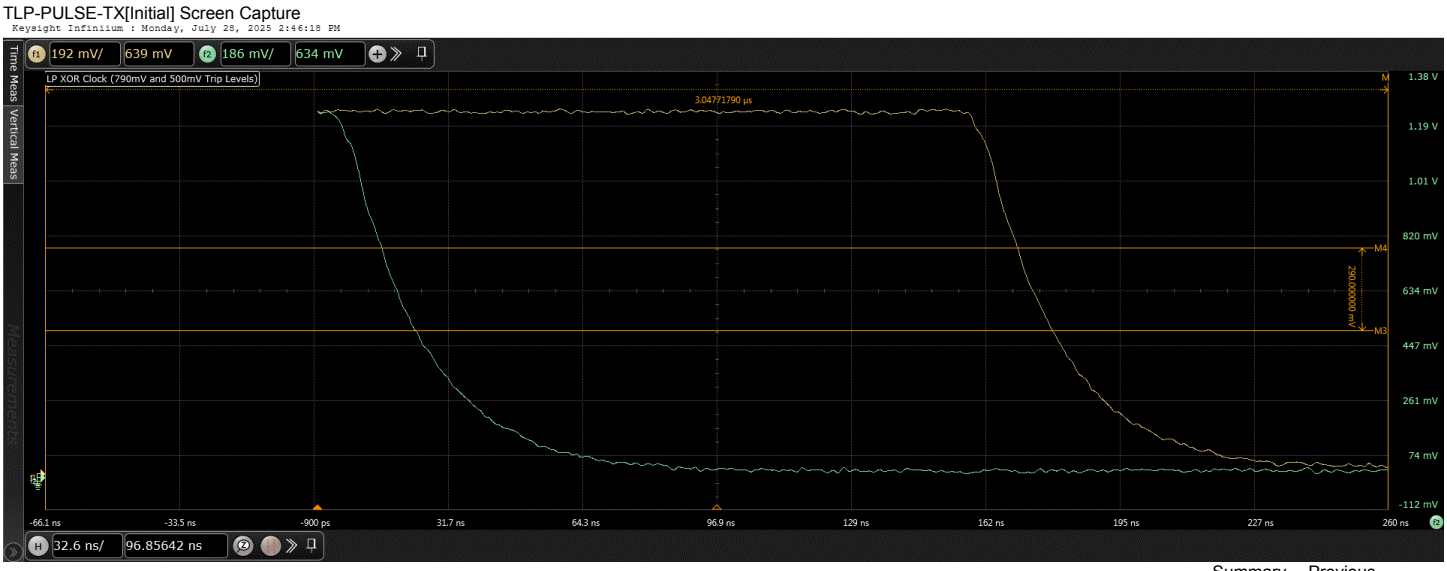
D-PHY Specification v2.1 Section 9.1.2 Table 28

Pulse width measurement for first pulse of the LP exclusive-OR clock.
Actual Value Measurement Name: 8271_Initial Pulse Width of LP XOR Clock(Worst)
Pass Limits: VALUE > 40.00 ns

Actual Value	Margin	Max Trip-level Threshold Voltage	Initial Pulse Width of LP XOR Clock@Max Trip-level Threshold Voltage
154.51 ns	286.3 %	790.00 mV	154.51 ns

Initial Pulse Width of LP XOR Clock@500mV	CLoad	DataLane	TLP-PULSE-TX[Initial] Screen Capture
154.61 ns	50pF	Lane0	(See image)

TLP-PULSE-TX[Initial] Screen Capture



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1.1.7 LP TX Period of LP TX Exclusive-OR Clock (TLP-PER-TX)

D-PHY Specification v2.1 Section 9.1.2 Table 28

Period of the LP exclusive-OR clock.
Actual Value Measurement Name: 828_Min Period of LP XOR Clock(Worst)
Pass Limits: VALUE > 90.00 ns

Actual Value	Margin	Max Trip-level Threshold Voltage	Min Period of LP XOR Clock@Max Trip-level Threshold Voltage
301.82 ns	235.4 %	790.00 mV	306.98 ns

Min Period of LP XOR Clock@500mV	CLoad	DataLane
301.82 ns	50pF	Lane0